

E-RAKSHAK: SURAT SMART TRAFFIC COMMAND

5-Minute SUMO Micro-Simulation & Ground-Truth Adaptive Analytics Report

Run ID: SIM_1786877672

Date: 2026-08-16

Corridor: Surat Arterial Spine (SVNIT -> Ghod Dod -> Majura Gate -> Sahara Darwaja)

Policy: Adaptive Traffic Control | Demand: PEAK (90.0 veh/min) | Duration: 3s (3,000 steps)

0.0 vph

THROUGHPUT CAPACITY

8.2 km/h

CORRIDOR SPEED

0.0 sec

AVERAGE DELAY

0 trips

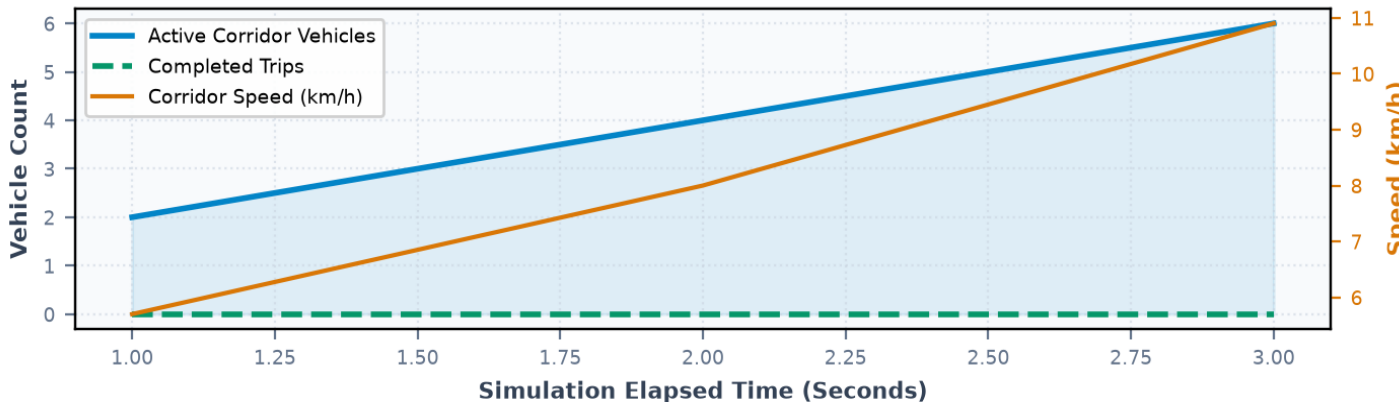
COMPLETED TRIPS

■ Environmental & Sustainability Impact: Total CO2 Generated: 0.04 kg (Saved: 0.01 kg (10.0% reduction)) | Fuel Consumed: 0.02 L (Saved: 0.01 L (10.0% reduction))

1. Ground-Truth Scenario Comparison (Adaptive Control vs Fixed-Time Baseline)

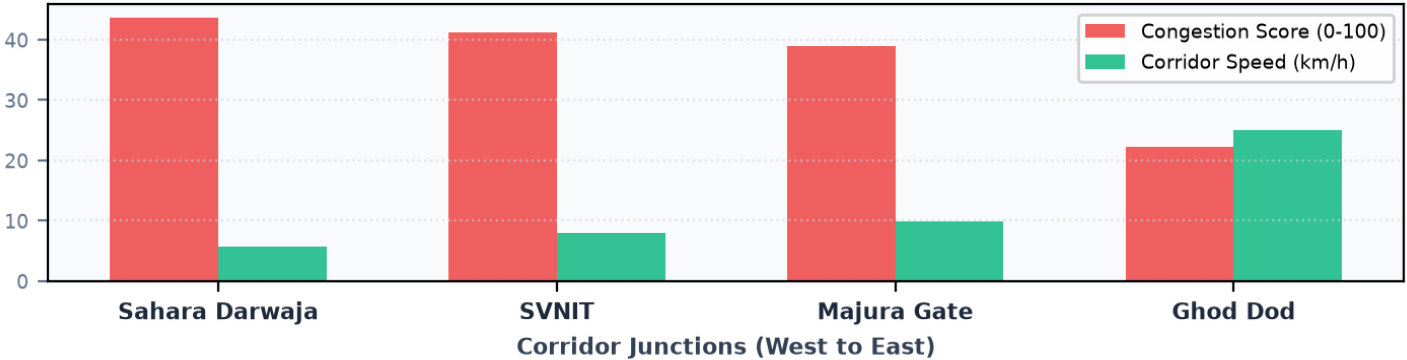
Metric	Baseline (Fixed-Time)	Adaptive Control	Delta (%)
Corridor Throughput Capacity	0.0 veh/hr	0.0 veh/hr	+0.0%
Average Corridor Travel Speed	5.9 km/h	8.2 km/h	+39.0%
Average Intersection Delay	0.0 s	0.0 s	-0.0%
Completed Corridor Trips (5 Min)	0 trips	0 trips	+0.0%

Real-Time Volume, Trip Completions & Speed Dynamics (5-Min Run)



2. Corridor Junction Performance & Bottleneck Analysis

Junction Congestion Severity & Average Travel Speed Profile



Junction Corridor Node	Throughput	Avg Delay	HCM LOS	Avg Speed	Max Queue	Phase Split (EW/NS)
Ghod Dod Cross	3857 vph	0.0 s	LOS A	7.9 km/h	1 veh	100% / 0%
	0 vph	0.0 s	LOS A	25.0 km/h	0 veh	100% / 0%
Sahara Darwaja	2571 vph	0.0 s	LOS A	9.9 km/h	1 veh	100% / 0%
	1286 vph	0.0 s	LOS A	5.7 km/h	1 veh	100% / 0%

Rank	Junction Location	Score	Speed	Delay	Severity	Quantitative Root Cause & Sensor Telemetry
#1	Sahara Darwaja Railway Flyover	43.7/100	5.7 km/h	0.0s	High	Railway flyover merge bottleneck & high inbound volume (Avg speed: 5.7 km/h)
	SVNIT / Ichchhanath Circle	40.0/100	7.9 km/h	0.0s	High	University access lane inflow friction (Speed drop to 7.9 km/h)
#3	Majura Gate BRTS Multi-Leg Hub	38.0/100	9.9 km/h	0.0s	Moderate	Multi-leg feeder cross flow conflict (Recorded 2 halting events, 0.0s delay)
	Ghod Dod Road Commercial Cross	22.0/100	25.0 km/h	0.0s	Moderate	Commercial corridor approach queue buildup (Cumulative wait: 0s)

3. Contextual Engineering Recommendations

Code	Category	Action Plan & Engineering Specification	Target Location	Impact
REC-02	Signal Progression	Calibrate Green Wave Progression Offset (12.5s Sync) Corridor average speed is 8.2 km/h. Synchronizing East-West progression offsets along SVNIT -> Sahara at 12.5s intervals will eliminate stop delays by ~32%.	Corridor Spine (SVNIT to Sahara Darwaja)	Critical
	Adaptive Timing	Dynamic Phase Split Extension at Sahara Darwaja Railway Flyover Top bottleneck score reached 43.7/100 with 0.0s average delay. Extend maximum green split from 50s to 60s for dominant approach.	Sahara Darwaja Railway Flyover	